AARON KAM

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EDUCATION

M.S. Molecular Biosciences and Bioengineering 2026 - Present
University of Hawai'i at Manoa
Department of Molecular Biosciences and Bioengineering

B.S. Information and Computer Science (Data Science Track) Cum Laude 2017 - 2022
University of Hawai'i at Manoa
Department of Information and Computer Science

B.S. Molecular Biosciences and Biotechnology Cum Laude 2017 - 2022
University of Hawai'i at Mānoa
Department of Molecular Biosciences and Bioengineering

SKILLS AND TECHNOLOGIES

- ***Bioinformatics:*** *Read preprocessing (fastp), file preparation (SAMtools, SRAtools), mapping and assembly (STAR, Salmon, Bowtie, Stringtie, Trinity), annotation (BLAST, enrichR, BUSCO)*
- ***Programming:*** *Python (OOP, Pandas, NumPy, SciPy, scikit-learn, matplotlib, Seaborn, Plotly Dash), SQL (Snowflake, MS SQL Server, SQLite), R (Bioinformatics packages)*
- ***Tech Stack:*** *SLURM; Bash; Git; Jupyter Notebook; R Markdown notebook; Anaconda; Singularity; Docker; Visual Studio Code; Snowflake; LaTeX;*
- ***Classical Machine Learning:*** *Classification (logistic regression, SVM, decision trees, random forests, neural network), Regression (LASSO, ridge, neural network).*
- ***ETL Pipelines:*** *PowerBI; Cerberus SFTP Server; SQL Server Management Studio; SQL Server Integration Services;*

PROFESSIONAL EXPERIENCE

University of Hawai'i at Mānoa *2022- Present*
Healthcare Data Analyst UH Telecommunications and Social Informatics Research Group

· *Data Engineering and ETL Development*

- *Worked on the planning and development of a healthcare claims data pipeline, from requirement gathering to integration, procurement, implementation, testing, and troubleshooting, leveraging commercial and in-house solutions to process 837 EDI files and supplemental documentation.*
- *Designed and implemented submission requirements and software using Python for duplicate detection, indexing, and validating completeness for claim files to a government agency.*
- *Utilized Python to analyze and profile member data from the CMNI Medicaid Eligibility and Enrollment Database, extracting insights into the coverage and demographic composition of the population.*

· *Data Science and Analytics*

- *Designed, implemented, and validated healthcare metric reports using Snowflake that provided usable insights for decision makers Hawai'i MedQUEST*

- *Performed data quality checks using Python to ensure accuracy and integrity of healthcare claims data spread across >50 Excel Spreadsheets for a client, identifying blind spots with operational compliance and improving claims processing workflows.*
- *Utilized Python to analyze and profile member data from the CMNI Medicaid Eligibility and Enrollment Database, extracting insights into the coverage and demographic composition of the population.*
- *Extracted and de-identified data from the CMNI Medicaid Eligibility and Enrollment Database using complex SQL queries, maintaining full compliance with HIPAA regulations.*

· *Business Analytics*

- *Investigated SQL databases from data partners to provide guidance for design and specification documents that were used for the implementation of public dashboards*
- *Collaborated directly with non-technical stakeholders to design business and technical requirements for the automatic data collection of healthcare claim submissions using various technologies such as SFTP, SSIS, Python, and SQL Server Database.*
- *Bridged the gap between existing but undocumented claim processing workflows into actionable guidelines, allowing immediate standardization and facilitating the planning of workflow enhancements for stakeholders.*

· *Technical Project Management*

- *Conducted code reviews and enforced coding standards to maintain best practices and ensure high-quality development of object-oriented Python scripts used in data pipelines.*
- *Mentored 4 student workers on data cleansing and the development of Power BI dashboards, resulting in the successful completion of multiple new dashboard projects.*

RESEARCH AND INTERNSHIPS

UH Telecommunications and Social Informatics Research Group *2021 - 2022*
Data Science Intern

· *Analyzed the CNMI Medicaid Eligibility and Enrollment database and its historic documentation for the development of materials relevant for planning, interim data validation, and population analysis.*
· *Created ETL processes for ingesting Excel spreadsheets containing claims data into a SQL database by linking them through Visual Studio and SSIS*

Dr. Jon-Paul Bingham's Lab *2025*
Research Volunteer

· *Planned and partially implemented a workflow and data pipeline to predict potential protein products and their characteristics from cone snail genomic data*
· *Mentored 2 students on the development the workflow that was presented at the CTAHR Summer Research Institute symposium.*

Dr. Dulal Borthakur's Lab *2018 - 2022*
Research Assistant

· *Performed RNA-Seq analysis and characterized proteins produced by Leucaena by utilizing Mana, UH's High Performance Computing cluster to assemble its predicted genes and analyzing their sequences*
· *Presented, summarized, and visualized the research findings at multiple symposiums for both experts and general audiences, with the results later used for publication in the scientific journal Frontiers in Plant Science.*

- ** Bageel, Ahmed M., Aaron Kam, and Dulal Borthakur. "Transcriptional analyses of genes related to fodder qualities in giant Leucaena under different stress environments." Frontiers in Plant Science 13 (2022): 885366.*

AWARDS AND HONORS

University of Hawaii at Manoa Dean's List *2017-2022*

UH Manoa Undergraduate Research Opportunity Fellow *2018*

IDeA Networks of Biomedical Research Excellence (INBRE) Fellow *2018*